

A Study of Implicit Quantile Networks for Risk-Sensitive Reinforcement Learning

Dipanshu Panda Sahaj Jayaram Shetty

Indian Institute of Technology Madras

- **Objective:** This study evaluates the Implicit Quantile Network (IQN) architecture for finding a CVaR-optimal policy
- **Core Comparison:** The study compares IQN with Quantile Regression DQN (QR-DQN) critics
- **Key Finding:** IQN provides superior stability, tighter confidence bounds on risk metrics, and consistent out-of-sample performance.
- **Validation:** A Successive Rejects bandit framework is used to validate IQN's risk estimation against empirical sample-based estimates.

Literature Review & Problem Statement

- **Context:** Traditional reinforcement learning maximizes expected returns, often neglecting extreme tail events present in financial markets.
- **Distributional RL:** Learning the entire return distribution allows agents to optimize risk measures such as Conditional Value at Risk (CVaR).
- **Problem:** While CVaR optimization is recognized as superior to mean-variance optimization, objectively verifying if distributional critics accurately represent tail risk remains unaddressed.
- **Goal:** Validate whether continuous quantile networks accurately align with real-world financial risk and lead to CVaR-optimal policies.

Architectural Comparison: IQN vs. QR-DQN

Quantile Regression DQN (QR-DQN)

- Approximates distributions by learning values for a fixed, discrete set of N quantiles.
- Struggles with continuous risk assessment and frequently misses value distributions between discrete points during extreme tail events.

Implicit Quantile Networks (IQN)

- Parameterizes the distribution continuously by taking the quantile fraction $\tau \sim U(0, 1)$ as an input.
- Samples any arbitrary risk level on the fly, providing a higher-resolution representation and smoother gradients.

The DSAC Algorithm

- **Framework:** Distributional Soft Actor-Critic (DSAC) replaces the scalar Q-critic in traditional SAC with a distributional critic $Z(x, a)$.
- **Risk Distortion:** The policy is trained to maximize a risk-distorted expected return rather than a simple expectation.
- **Optimization Objective:** The actor optimizes the CVaR objective at a risk level $\alpha = 0.25$.
- **Mechanism:** This objective signals the policy to elevate the lower bounds of the predicted return distribution, enforcing risk-averse behavior.

DSAC Algorithm - Pseudocode

Algorithm 1 DSAC Update

- 1: Input: $s, a, r, s', \gamma \in (0, 1)$, Parameter: N
 - 2: Generate quantile fractions $\tau_i, \hat{\tau}_j$
 - 3: **for** $i = 0$ to $N - 1$ **do**
 - 4: Compute target quantiles: $y_i = \min_{k=1,2} Z_{\hat{\tau}_i}(s', a'; \bar{\theta}_k)$
 - 5: **for** $j = 0$ to $N - 1$ **do**
 - 6: Compute TD error: $\delta_{ij}^k = r + \gamma[y_i - \alpha \log \pi(a'|s')] - Z_{\hat{\tau}_j}(s, a; \theta_k)$
 - 7: **end for**
 - 8: **end for**
 - 9: Update Critic networks θ_k using quantile Huber loss $\mathcal{J}_Z(\theta_k)$
 - 10: Compute expected Q-value $Q(s, \tilde{a})$ from distribution
 - 11: Update Actor network ϕ using entropy-regularized loss $\mathcal{J}_\pi(\phi)$
 - 12: Soft update target networks $\bar{\theta}_k$ and $\bar{\phi}$
-

Environment 1: Synthetic MDP

- **Structure:** A 4-state chain starting at initial state x_0 and ending at terminal state x_3
- **Actions & Rewards:** Each state offers a choice between two actions with Gaussian distributions:
 - **Action a_0 (Risk-Neutral Choice):** Immediate reward $r \sim \mathcal{N}(1, 1)$.
Higher expected return, but higher variance/risk.
 - **Action a_1 (Risk-Averse Choice):** Immediate reward $r \sim \mathcal{N}(0.8, 0.4^2)$.
Lower expected return, but significantly tighter distribution.
- **Advantage:** The environment possesses an analytical ground truth where CVaR reduces to a closed-form integral.
- **Dynamics:** The risk-neutral optimal action yields a higher mean, but the risk-averse optimal switches to the lower variance action for risk levels $\alpha \leq 0.62$.

Environment 2: FinRL Stock Trading

- **Stocks:** Evaluated on five major large-cap stocks (AAPL, MSFT, GOOGL, AMZN, NVDA).
- **Dataset:** Training from 2016–2021; Testing/Backtest from 2022–2023.
- **Market Regime:** The test set captures the 2022 tech drawdown and 2023 recovery, offering a robust test for risk constraints.
- **State & Action:** High-dimensional state space (prices, technicals, VIX); continuous action space for portfolio weights.
- **Reward:** Immediate change in total portfolio value, scaled by 10^{-2} to stabilize gradients.

Methodology: Synthetic MDP Training

- **CVaR Control Approaches:**

- ① *Markov CVaR Bellman*: The standard risk-sensitive IQN where the action is selected as $\arg \max CVaR_\alpha(Z(s, a))$, averaging the bottom- α quantiles.
- ② *Malik W_θ* : Tracks a running threshold s_t per episode ($s_{t+1} = (s_t - r_t)/\gamma$, $s_0 = q_\alpha(Z(x_0, \pi(x_0)))$) and uses an auxiliary value $W_\theta(s, a)$ that conditions on both the current state and the threshold. Action selection is $\arg \max_a W_\theta(s, a)$.

- **Two-Phase Training:**

- *Phase 1*: Train a CVaR-greedy policy.
- *Phase 2*: Freeze the actor and train a fresh IQN critic purely for distributional policy evaluation.

- **Scope**: Evaluated across multiple α levels on 10 random seeds.

Methodology: DSAC Training Pipeline (FinRL)

- Models are trained across 5 random seeds to ensure significance.
- **Initialization:** Network weights and environment dynamics are reset at the beginning of each run.
- **Training:** Executes for 200 epochs, carefully balancing exploration using MDP Path Collectors and exploitation.
- **Evaluation:** Post-training, a deterministic evaluation policy backtests the agent on the unseen 2022-2023 test dataset.

Bandit Framework for CVaR Validation

- **Concept:** A multi-armed bandit algorithm is employed to validate IQN's internal risk estimation accuracy.
- **Algorithm:** Uses Successive Rejects (SR) to eliminate suboptimal arms.
- **Each arm** represents one of the last ten iterates (policies) of a DSAC algorithm run.
- **The Comparison:** The bandit compares two estimators:
 - ① *IQN Native Estimation:* Analytical CVaR with frozen critic networks.
 - ② *Empirical Sample-Based Estimation:* CVaR computed with noisy rollouts.

Bandit Framework (Pseudocode)

Algorithm 2 Successive Rejects (SR) Algorithm

- 1: **Input:** Number of arms K , total budget of rounds n
- 2: **Initialize:**
 - Let $A_1 = \{1, \dots, K\}$ ▷ Set of all active arms at the start
 - $\overline{\log(K)} = \frac{1}{2} + \sum_{i=2}^K \frac{1}{i}$ ▷ Harmonic-like normalization constant
 - $n_0 = 0$
 - For $k \in \{1, \dots, K-1\}$, $n_k = \left\lceil \frac{1}{\overline{\log(K)}} \frac{n-K}{K+1-k} \right\rceil$ ▷ Compute cumulative pulls per arm up to phase k
- 3: **for** each phase $k = 1, 2, \dots, K-1$ **do**
- 4: **for** each $i \in A_k$ **do**
- 5: Select arm i for $n_k - n_{k-1}$ rounds ▷ Pull each active arm for the allotted budget of this phase
- 6: **end for**
- 7: Let $A_{k+1} = A_k \setminus \arg \min_{i \in A_k} \hat{X}_{i, n_k}$ ▷ Dismiss the arm with the lowest empirical mean
- 8: **end for**
- 9: **Output:** Let J_n be the unique element of A_K ▷ The single surviving arm after $K-1$ eliminations
- 10: **Return** J_n

Stochastic Environment for Bandit Validation

- **Problem:** Deterministic backtests produce identical trajectories, making sample-based variance estimation trivial and unrealistic.
- **Solution:** Developed a `StochasticStockTradingEnv` wrapper to introduce realistic uncertainty.
- **Aleatoric Uncertainty Injection:**
 - Injects Gaussian noise ($\sigma = 0.001$) into both state observations and step rewards.
 - Forces the estimators to handle environment stochasticity during policy evaluation.
- **The Goal:** Create a fair testing ground to compare how IQN's internal distribution learning handles noise versus traditional empirical rollouts.

Evaluation Metrics

DSAC Performance Metrics (FinRL)

- **Total Return (%)**: Cumulative profit over the backtest.
- **Sharpe Ratio**: Risk-adjusted return.
- **Max Drawdown (%)**: Largest peak-to-trough drop.
- **CVaR 5% (%)**: Expected loss in the worst 5% of days.

Bandit Validation Metrics

- **Oracle**: Ground truth CVaR established via 500 Monte Carlo episodes.
- **Selection Accuracy (%)**: Frequency of successfully identifying the exact Oracle-best arm.
- **Mean CVaR Regret**: Average true CVaR difference between the optimal arm and the selected arm.

Results: Synthetic MDP

- Both Markov CVaR and Malik W_θ successfully tracked the trend that CVaR increases with α .
- Tail Error:** IQN tends to overestimate CVaR in extreme tails ($\alpha \leq 0.25$).
- Baseline Comparison:** A standard risk-neutral baseline is drastically worse at tail estimation, lacking data on rare low-return events.
- MAE:** At $\alpha = 0.10$, risk-neutral MAE is 0.756 compared to Markov CVaR MAE of 0.270.

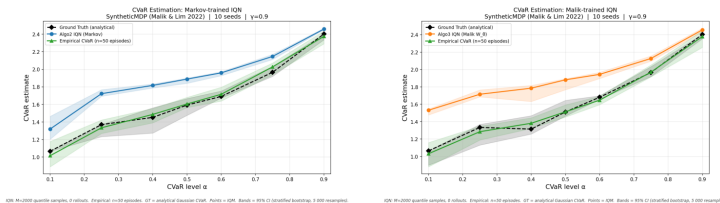


Figure: IQN estimate vs. analytical ground truth as a function of α for both the Markov CVaR Bellman and Malik W_θ algorithms. The overestimation is most

Results: FinRL (Return & Volatility)

Metric	DSAC + IQN	DSAC + QR-DQN
Total Return (%)	28.154 ± 8.778	20.600 ± 13.309
Sharpe Ratio	0.558 ± 0.107	0.462 ± 0.181
Max Drawdown (%)	-34.338 ± 1.151	-36.209 ± 5.523
CVaR 5% (%)	-4.084 ± 0.073	-4.226 ± 0.389
VaR 5% (%)	-2.975 ± 0.104	-3.054 ± 0.315
Calmar Ratio	0.824 ± 0.263	0.619 ± 0.419
Win Rate (%)	50.240 ± 0.463	51.040 ± 0.697

Table: FinRL Backtest Performance Summary (Mean $\pm$ Std across 5 seeds).

- **Verdict:** IQN achieved a higher mean total return (28.15% vs. 20.60%) and Sharpe ratio (0.558 vs. 0.462) over QR-DQN.
- **Consistency:** IQN achieved these returns with substantially lower variance.

Results: FinRL (Return & Volatility)

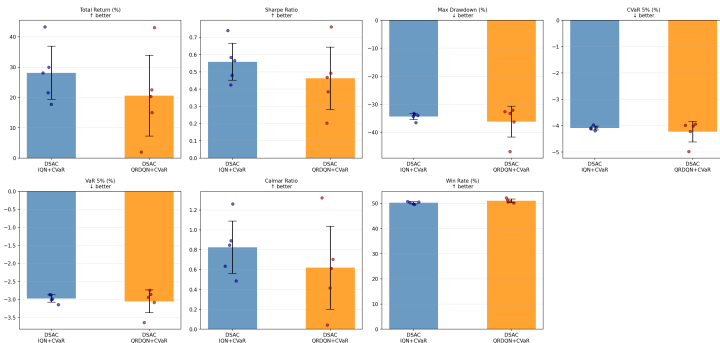
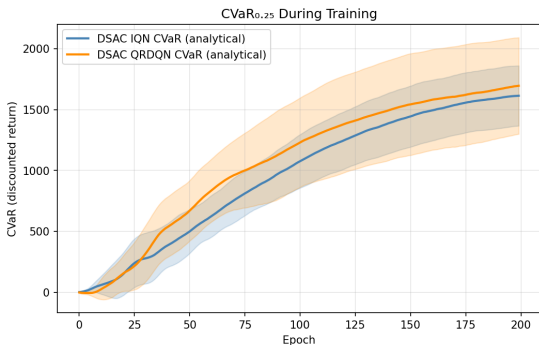


Figure: Backtest performance distributions across 5 seeds.

- **QR-DQN Flaws:** QR-DQN occasionally hits high returns but frequently collapses, leading to a massive standard deviation ($\pm 13.3\%$).

Results: FinRL (Tail-Risk Management)



- IQN consistently protected the portfolio's downside.
- **CVaR Variance:** The standard deviation for IQN's CVaR 5% was just $\pm 0.073\%$, outperforming QR-DQN's $\pm 0.389\%$.
- **Drawdowns:** IQN's Maximum Drawdown was tightly bounded at $-34.338 \pm 1.151\%$.

Results: FinRL (Training Dynamics)

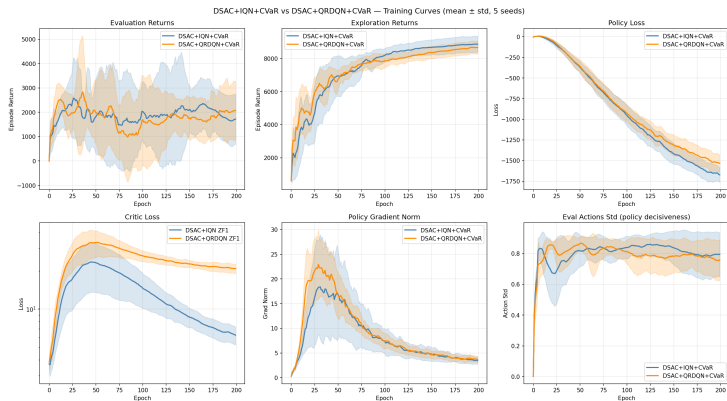


Figure: Training dynamics over 200 epochs.

Results: FinRL (Training Dynamics)

- **Convergence:** IQN's Critic Loss converged much more smoothly and to a lower baseline than QR-DQN's.
- **Gradient Signals:** Because IQN samples continuous fractions $\tau \sim U(0, 1)$, gradient updates are richer and provide a higher-resolution understanding of value distributions.
- **Decisiveness:** IQN's policy decisiveness remained steady and confident throughout training.
- **Instability:** QR-DQN showed higher target variability, causing its policy to fluctuate and become less decisive.

Results: Bandit Performance

- **Superior Selection:** The IQN-driven bandit outperformed the empirical sample-based bandit in identifying CVaR-optimal policies.
- **Regret Metric:** IQN selection had a superior Mean CVaR Regret of 0.0486%, compared to the sample-based estimator's 0.0791%.
- **Noise Isolation:** By evaluating analytical CVaR from frozen critics, IQN effectively isolated epistemic policy uncertainty from aleatoric market noise.
- **Robustness:** Empirical estimators frequently eliminated strong arms prematurely due to unlucky stochastic rollouts.

Conclusion

- IQN is validated as a state-of-the-art architecture for policy evaluation in financial environments.
- **Structural Advantage:** Continuous parameterization effectively avoids the blind spots inherent to discrete approximations like QR-DQN.
- **Performance:** DSAC+IQN delivers superior total returns while maintaining reliability required for real-world algorithmic trading.
- **Final Verdict:** IQN consistently and accurately mitigates worst-case losses, proving to be a useful framework for risk-aware asset management.